

Case study of Service models in cloud computing

- *Background:*
- A company named "TechSolutions" aims to develop and deploy a web application for project management. They decide to leverage Cloud Platform, for their project.

Infrastructure as a Service (IaaS) with Google Compute Engine:

Component	Description
Service Used	Google Compute Engine (IaaS)
Functionality	Provisioning virtual machines, managing own OS/apps.
Advantages	Full control over infrastructure, flexibility.
Challenges	Requires management of underlying OS and applications.

Platform as a Service (PaaS) with Google App Engine:

Component	Description
Service Used	Google App Engine (PaaS)
Functionality	Focus on app development, Google manages infrastructure.
Advantages	Streamlined development, automatic scaling, multiple languages support.
Challenges	Limited flexibility compared to managing own infrastructure.

Software as a Service (SaaS) Integration:

Component	Description
Services Used	Google Cloud Storage (SaaS), Google Cloud SQL (SaaS), Google Cloud Pub/Sub (SaaS) :- pub/sub (real-time messaging service using a publish/subscribe model, allowing applications to send and receive messages)
Functionality	Storage, managed database, messaging service.
Advantages	Ready-to-use services, minimal management required.
Challenges	Data integration complexity, dependency on external services.

Infrastructure as a Service (IaaS) with Microsoft Azure:

Component	Description
Service Used	Azure Virtual Machines (IaaS)
Functionality	Provisioning VMs, managing own OS/apps.
Advantages	Control over infrastructure, scalability, multiple OS support.
Challenges	Requires management of underlying OS and applications.

Platform as a Service (PaaS) with Microsoft Azure:

Component	Description
Service Used	Azure App Service (PaaS)
Functionality	Focus on app development, Azure manages infrastructure.
Advantages	Streamlined development, automatic scaling, deployment slots.
Challenges	Limited to specific languages/frameworks, less control over infrastructure.

Software as a Service (SaaS) Integration:	
Component	Description
Services Used	Azure Blob Storage (SaaS), Azure SQL Database (SaaS), Azure Service Bus:-cloud messaging service that facilitates reliable communication between applications and services.
Functionality	Storage, managed database, messaging service.
Advantages	Ready-to-use services, high availability, security features.
Challenges	Data integration complexity, vendor lock-in concerns.

Infrastructure as a Service (IaaS) with Amazon EC2:	
Component	Description
Service Used	Amazon EC2 (IaaS)
Functionality	Provisioning VMs, managing own OS/apps.
Advantages	Scalability, choice of OS, various instance types.
Challenges	Requires management of underlying OS and applications.

Platform as a Service (PaaS) with AWS Elastic Beanstalk:	
Component	Description
Service Used	AWS Elastic Beanstalk (PaaS)
Functionality	Focus on app development, AWS manages infrastructure.
Advantages	Easy deployment, automatic scaling, multiple language support.
Challenges	Limited control over underlying infrastructure.

Software as a Service (SaaS) Integration:	
Component	Description
Services Used	Amazon S3, Amazon RDS, Amazon SQS (allows users to send, store, and retrieve messages between applications.)
Functionality	Storage, managed database, messaging service.
Advantages	Scalability, durability, minimal maintenance.
Challenges	Data integration complexity, security considerations.

Infrastructure as a Service (IaaS) with Eucalyptus:

- Eucalyptus is an acronym for **Elastic Utility Computing Architecture for Linking Your Programs To Useful Systems**.
- It's an open-source software platform that allows organizations to build their own private or hybrid cloud environments using their existing hardware infrastructure.

Component	Description
Service Used	Eucalyptus (IaaS)
Functionality	Provisioning VMs, managing own OS/apps.
Advantages	Private cloud deployment, integration with existing infrastructure.
Challenges	Requires management of underlying OS and applications.

Amazon EC2 Case Study

Amazon EC2 (Elastic Compute Cloud) is a web service offered by Amazon that provides resizable compute capacity in the cloud. It is designed to make web-scale cloud computing easier for developers. One of the biggest advantages of using Amazon EC2 is its ability to quickly scale up or down as needed, which makes it an ideal choice for businesses with fluctuating compute needs.

One real-life example of a company using Amazon EC2 is Airbnb. In 2011, Airbnb experienced a significant increase in traffic due to a surge in demand for accommodation during the South by Southwest (SXSW) festival. This sudden increase in traffic caused the company's website to crash, which led to a loss of potential revenue.

To prevent this from happening again, Airbnb decided to migrate its infrastructure to the cloud and chose Amazon EC2 as its primary compute service. With Amazon EC2, Airbnb was able to quickly scale up its infrastructure to handle the increased traffic during peak times, such as during major events or holidays.

Additionally, Amazon EC2 allowed Airbnb to save costs by only paying for the compute capacity they actually needed at any given time. This allowed the company to avoid over-provisioning, which is a common issue when using on-premises infrastructure.

In 2015, Airbnb experienced another surge in demand during the New Year's Eve holiday season. Thanks to Amazon EC2, the company was able to handle over 2 million guests booking on their platform without any major issues.

Overall, Amazon EC2 has provided Airbnb with the flexibility and scalability it needs to handle unexpected spikes in traffic and the ability to manage costs efficiently. This has allowed the company to focus on providing the best possible experience for its users while also ensuring the stability and reliability of its infrastructure.

What led Airbnb to choose Amazon EC2 as its primary compute service?

A: Airbnb chose Amazon EC2 as its primary compute service because of its ability to quickly scale up or down as needed, which makes it an ideal choice for businesses with fluctuating compute needs. Additionally, Amazon EC2 allowed Airbnb to save costs by only paying for the compute capacity they actually needed at any given time, which helped the company avoid over-provisioning.

Q: How did Amazon EC2 help Airbnb during the New Year's Eve holiday season?

A: During the New Year's Eve holiday season, Airbnb experienced a surge in demand for its platform. Thanks to Amazon EC2, the company was able to handle over 2 million guests booking on their platform without any major issues. Amazon EC2 provided Airbnb with the flexibility and scalability it needed to handle unexpected spikes in traffic and ensure the stability and reliability of its infrastructure.

Q: What are some of the benefits of using Amazon EC2 for businesses?

A: Some of the benefits of using Amazon EC2 for businesses include the ability to quickly scale up or down as needed, efficient cost management by paying only for the compute capacity they actually need at any given time, and the ability to focus on providing the best possible experience for users while ensuring the stability and reliability of their infrastructure. Additionally, Amazon EC2 provides businesses with a secure and reliable platform for their computing needs.

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Google Cloud Case Study

One of the most popular cloud computing services available today is Google Cloud, which is offered by Google as a suite of cloud computing services. Google Cloud provides a range of services including storage, networking, data analytics, and machine learning, among others. Here's a case study on how Google Cloud has been used by one company:

The Home Depot is a US-based home improvement retailer with more than 2,200 stores across North America. In 2017, The Home Depot made a strategic decision to shift its infrastructure to the cloud and chose Google Cloud as its primary cloud provider.

With the help of Google Cloud, The Home Depot was able to modernize its IT infrastructure and improve the scalability and reliability of its systems. Google Cloud provided The Home Depot with a range of tools and services to support its digital transformation, including:

Compute Engine: The Home Depot used Compute Engine to quickly spin up virtual machines (VMs) to support its applications and services.

Cloud Storage: The Home Depot used Cloud Storage to store its data in a secure and reliable manner.

BigQuery: The Home Depot used BigQuery to analyze and gain insights from its data, which helped the company improve its operations and make data-driven decisions.

Kubernetes: The Home Depot used Kubernetes to manage its containerized applications and services, which helped the company improve its agility and responsiveness.

By using Google Cloud, The Home Depot was able to improve the reliability and scalability of its IT infrastructure, reduce its IT costs, and enable faster innovation. For example, The Home Depot was able to reduce the time required to set up a new application from several weeks to just a few hours, which enabled faster deployment of applications and services.

Overall, Google Cloud has provided The Home Depot with the tools and services it needs to support its digital transformation and improve its operations. The Home Depot has been able to leverage the scalability, reliability, and agility of Google Cloud to support its growth and provide its customers with a better experience.

Q: What led The Home Depot to choose Google Cloud as its primary cloud provider?

A: The Home Depot made a strategic decision to shift its infrastructure to the cloud in 2017 to modernize its IT infrastructure, improve the scalability and reliability of its systems, and reduce its IT costs. The company chose Google Cloud as its primary cloud provider because it provided a range of tools and services to support its digital transformation, including Compute Engine, Cloud Storage, BigQuery, and Kubernetes.

Q: How did Google Cloud help The Home Depot reduce the time required to set up a new VM?

A: Google Cloud provided The Home Depot with Compute Engine, which allowed the company to quickly spin up virtual machines (VMs) to support its applications and services. By using Compute Engine, The Home Depot was able to reduce the time required to set up a new VM from several weeks to just a few hours, which enabled faster deployment of new applications and services.

Q: What benefits did The Home Depot gain from using Google Cloud?

A: The Home Depot gained several benefits from using Google Cloud, including improved reliability and scalability of its IT infrastructure, reduced IT costs, and faster innovation. With the help of Google Cloud, The Home Depot was able to modernize its IT infrastructure, improve its operations, and make data-driven decisions using BigQuery. Additionally, The Home Depot was able to manage its containerized applications and services more effectively using Kubernetes. Overall, Google Cloud provided The Home Depot with the tools and services it needed to support its digital transformation and improve its operations.

Case Study: Microsoft Azure and CrowdStrike Incident

Background

In July 2023, a widespread IT outage occurred due to a faulty update released by CrowdStrike, a cybersecurity company. The issue primarily affected Microsoft Windows systems globally, causing disruptions in banking, airlines, healthcare, and government services. The root cause was an update that led to blue screen errors (BSOD) and system crashes, severely impacting enterprises relying on Azure's cloud infrastructure.

Key Facts of the Incident

- CrowdStrike deployed a faulty security update to its Falcon sensor software.
- The update triggered a crash in Windows systems, leading to widespread downtime.
- Many enterprises using Microsoft Azure experienced outages since their cloud-based virtual machines were also running Windows.
- Microsoft worked with CrowdStrike to roll back the update and provided guidance to affected users.

Case Study Questions and Answers

1. What was the primary cause of the incident?

Answer:

The incident was caused by a faulty update in the CrowdStrike Falcon security software, which led to Windows systems crashing with blue screen errors (BSOD). This affected both on-premise and cloud-based Windows instances, including those on Microsoft Azure.

2. How did the issue impact businesses and cloud users?

Answer:

The faulty update led to system failures in multiple industries, including:

- **Banking & Finance:** ATM services and online banking were disrupted.
- **Airlines & Transportation:** Flight operations faced delays and cancellations.
- **Healthcare:** Hospitals and medical services faced IT disruptions.
- **Government & Enterprises:** Cloud-based services experienced downtime, affecting productivity.

3. Why was Microsoft Azure particularly affected by this incident?

Answer:

Microsoft Azure provides cloud-based virtual machines that run on Windows. Since many enterprises use CrowdStrike Falcon for security, the faulty update impacted a significant number of virtual machines in the Azure environment, causing widespread service disruptions.

How did Microsoft and CrowdStrike respond to the crisis?

Answer:

- **Microsoft:** Issued emergency guidelines on how to prevent further system crashes, including rolling back updates and restarting machines in safe mode.
- **CrowdStrike:** Quickly identified the problem, provided patches, and worked with Microsoft to mitigate the damage. They also took responsibility for the error and communicated transparently with affected customers.

5. What lessons can be learned from this incident?

Answer:

- **Importance of rigorous update testing:** Software updates should undergo extensive testing before deployment.
- **Need for rollback mechanisms:** Cloud providers must have quick rollback strategies for critical security updates.
- **Effective communication during crises:** Vendors must promptly inform customers and provide mitigation steps.
- **Redundancy and backup strategies:** Enterprises should have contingency plans, such as backup systems, to minimize downtime during unexpected failures.